

EventLens

The Intelligent Event Media Management Platform

AI-Powered Discovery | Real Time Collaboration | Smart Infrastructure

The "Post Event" Media Nightmare

Organizations and Clubs host many incredible events, but managing the aftermath is painful.

- **Extreme Fragmentation:** Thousands of photos and videos get scattered across Drives and local folders.
- **The Search Penalty:** Attendees waste hours scrolling to find the 3 photos they appear in.
- **Hardware Limitations:** Serving massive raw videos constantly crashes web servers.
- **Privacy Liabilities:** Unrestricted access links expose private club photos.

The Solution

A centralized ecosystem engineered for modern event teams.

EventLens replaces scattered drives with a singular, intelligent hub that categorizes, optimizes, protects, and delivers event memories flawlessly.

- **For Organizers:** Effortless bulk uploading, AI tagging, and instant analytics.
- **For Attendees:** Instant AI facial discovery, QR-based sharing, and seamless social engagement.

Media Engine & Event Architecture

Built to Scale. Engineered for Delivery Speed.

Media Management

- **Multipart Chunking:** Safely streams massive videos.
- **Dynamic WebP:** Slashes bandwidth by 80%.
- **Duplicate Image Detection:** Auto rejects duplicate uploads.
- **Offline Caching:** Media loads instantly, even offline.

Event Architecture

- **Nested Albums:** Organize events hierarchically.
- **Collaborative Albums:** Multiple photographers can upload seamlessly.
- **QR-Based Sharing:** Scan to instantly share or join an event album without accounts!

"Find My Photos" & Advanced Search

The End of Manual Photo Searching

Facial Recognition AI

- **Edge Extraction:** Browser extracts a mathematical multi dimensional vector array.
- **Math Engine:** Postgres backend scans thousands of photos in seconds.
- **The Result:** A gallery of every photo you appear in, generated in seconds.

Advanced Metadata Search

Filter media instantly by:

- Event Name
- AI-Generated Scene Tags
- Upload Date
- Specific Photographer Name

Security, Moderation & Protection

- **Streamlined RBAC:** Admin, Photographer, Member, and Viewer roles managed via a modern permissive session check.
- **Automated Image Moderation:** AWS AI automatically scans and flags inappropriate content during upload.
- **Dynamic Watermarking:** Automatically burns Club & Event logos onto media downloaded by standard viewers to protect intellectual property.
- **Data Integrity:** Strict PostgreSQL foreign key constraints ensure ghost records and broken relations are impossible.

Real Time Social & Global Analytics

Turning static folders into interactive experiences

Real Time Social Layer

- **Server Sent Events (SSE):** Likes, comments, and uploads update globally in real time.
- **Global Activity Feeds:** A centralized chronological timeline tracking all interactions across an event.

Analytics Dashboard

Comprehensive Admin tracking:

- Storage utilization insights
- User engagement (likes/comments mapping)
- Event growth and upload velocity metrics

Tech Stack

- **Frontend:** Next.js (Turbopack) & React 19 for instantaneous page loads. Styled with Tailwind CSS and Framer Motion.
- **Backend:** Node.js & Express.js API Gateway handling heavy stream buffers.
- **Database:** PostgreSQL managed securely via Prisma ORM.
- **AI & Cloud:** `face-api.js` (for facial recognition), Cloudinary SDK, AWS Rekognition.

Key Learnings

- **Real-Time Features:** Building live "likes" and comments demonstrated how to instantly sync data across multiple users at the same time.
- **Smart AI Processing:** Moving the heavy AI facial scanning to the user's browser improves app speed and protects user privacy.
- **Access Control Systems:** Building a strict role-based system highlighted the importance of keeping private event media secure from unauthorized viewers.
- **Cloud Integrations:** Securely connecting to external cloud services for storage and moderation requires careful management of API keys and data streams.

Future Goals

What more can be done for the EventLens Platform?

- **AI Highlight Reels:** Automatically cutting together the best moments and most tagged users from an event into an auto-generated, shareable video.
- **Voice-Powered Retrieval:** Natural language querying ("Show me photos of the robotics team from yesterday afternoon").
- **Native Mobile App:** Expanding the Next.js foundation into dedicated iOS and Android experiences.

Thank You
